

Padding and Stride

1. Output size shrinks

For input size $N \times N$ and kernel $F \times F$:

$$\text{Output} = (N - F + 1)$$

Example:

5x5 input with 3x3 filter $\rightarrow$ 3x3 output

After multiple layers:

$5 \times 5 \rightarrow 3 \times 3 \rightarrow 1 \times 1 \rightarrow \text{disappears}$

So deep networks cannot be built effectively without padding.

2. Loss of edge information

- Filters cannot fully operate on border pixels
- Corner and edge pixels are used fewer times

Effect:

- Important features near edges are lost

3. Unequal contribution of pixels

- Center pixels are used in many convolution operations
- Edge pixels are used very few times

Effect:

- Model becomes biased toward central regions

4. Rapid loss of spatial information

- Feature maps shrink quickly
- Fine details are lost early

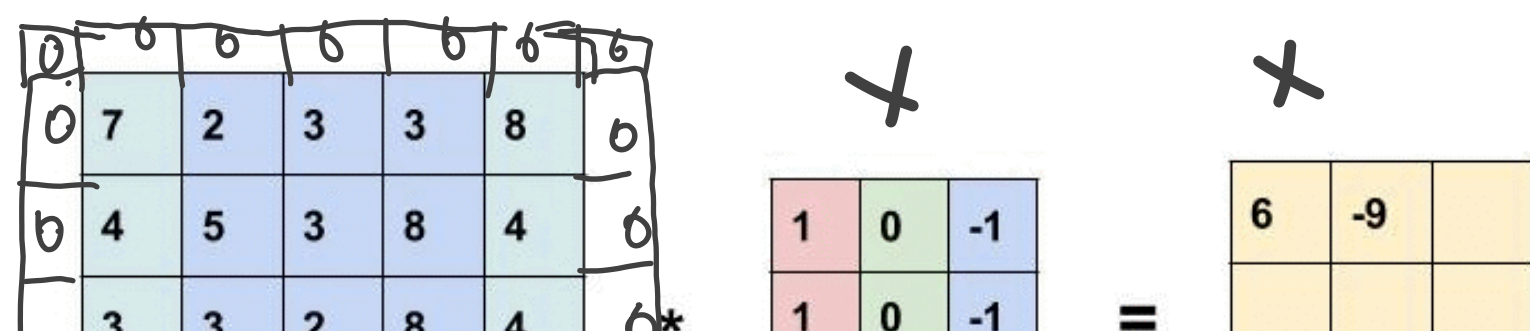
Effect:

- Poor performance in tasks requiring spatial details (e.g., detection, segmentation)

5. Feature maps collapse in deep networks

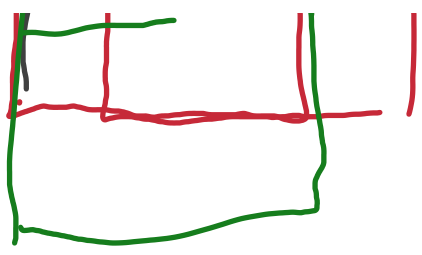
- After a few layers, spatial dimensions become too small
- Eventually no meaningful spatial structure remains

Without padding, convolution reduces spatial dimensions at every layer and underutilizes edge pixels, leading to information loss and limiting the depth and effectiveness of CNNs.



$$\begin{aligned} &2x_1 + 5x_1 + 3x_1 + \\ &3x_0 + 3x_0 + 2x_0 + \\ &3x_{-1} + 8x_{-1} + 8x_{-1} \\ &= -9 \end{aligned}$$

24

 7×7

373

padding
+
Stride

$$\left[\frac{x+2p-f+g}{s_i} \right]$$

Impact of increasing stride (S) in convolution

Formula:

$$\text{Output size} = (N + 2P - F) / S + 1$$

- ✓ 1. Output size becomes smaller

- As stride (S) increases, output size decreases
- Because we divide by S

Example:

Stride = 1 $\rightarrow$ large output

Stride = 2 $\rightarrow$ smaller output

2. Downsampling happens

- Increasing stride reduces spatial resolution
- Similar to pooling

3. Information loss increases

- Filter skips positions
- Some pixels are never used

4. Computation becomes faster

- Fewer operations are needed
- Improves speed

5. Less detail in feature maps

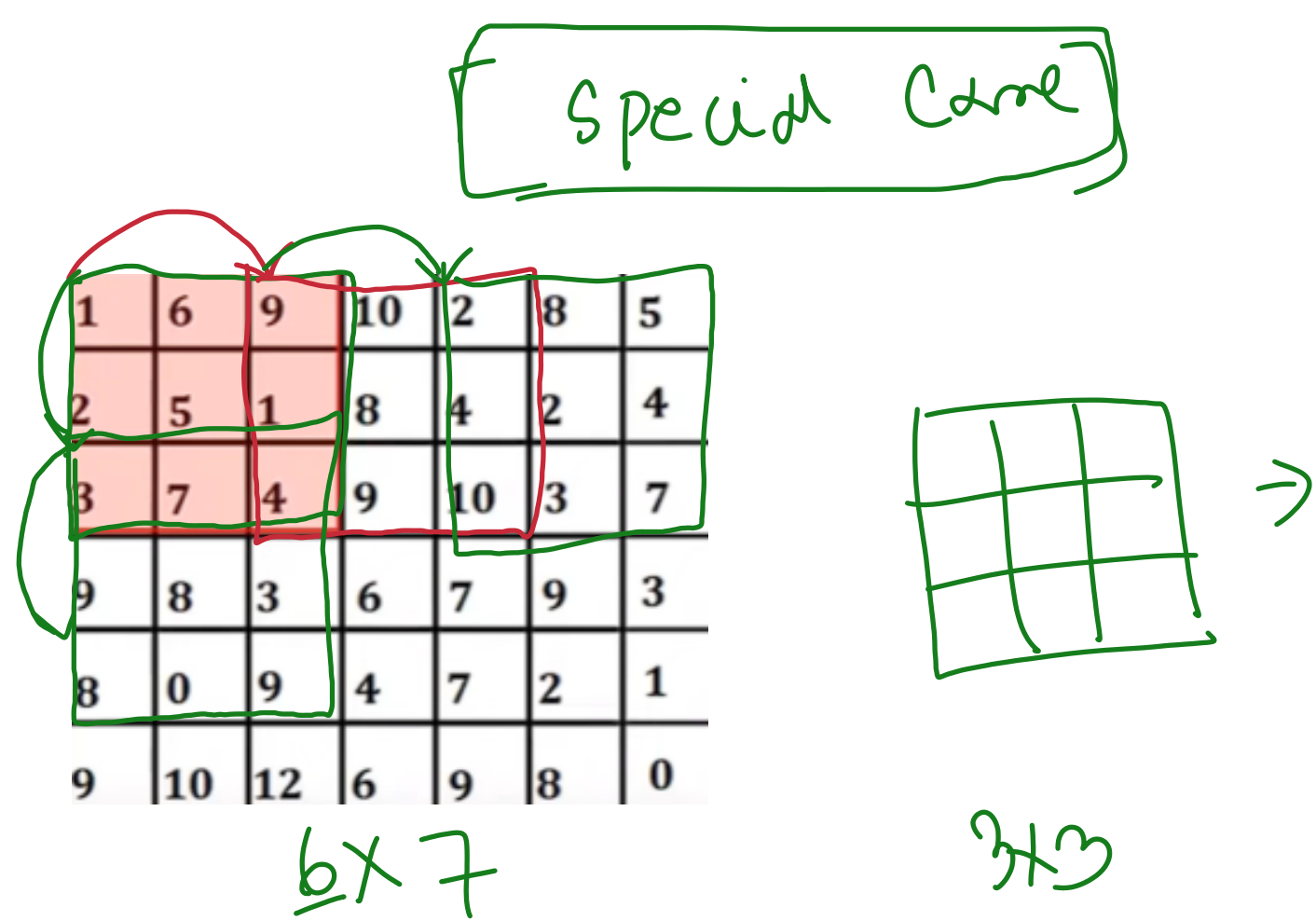
- Fine details are lost
- Only important patterns remain

- 6. Can reduce accuracy (if too large)

- Too much stride $\rightarrow$ too much information loss

Summary

Output size = $(N + 2P - F) / S + 1$ shows that increasing stride reduces output size, making the model faster but causing loss of information.



eg: $\frac{n-f}{5} + 1$

$$4n-6 \rightarrow \frac{6-3}{2} + 1 \rightarrow \frac{3}{2} + 1$$

$\rightarrow 1.5 + 1$

$\rightarrow 1+1=2$

$$\underline{\underline{n=7}} \quad \frac{7-9}{2} + 1 = \frac{4}{2} + 1 = 3$$

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